

Ahmed Thabet

Engineering Manager — Distributed Cloud Infrastructure & Platform Engineering

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Engineering leader with 10+ years building decentralized cloud infrastructure, custom operating systems, and developer platforms — from bare-metal OS design to Kubernetes orchestration at scale. Currently managing multiple teams delivering a peer-to-peer cloud that spans thousands of nodes across 60+ countries.

Experience

Engineering Manager, Codescalers, Jan 2023 - Present

- Manage multiple engineering teams building the [ThreeFold Grid](#), a decentralized P2P cloud spanning thousands of nodes across 60+ countries.
- Own technical strategy, architecture, roadmap, and delivery across the full stack — a custom Linux-based OS, multi-language SDKs, a Terraform provider, and web dashboards.
- Run hiring, onboarding, and mentoring; set engineering standards for code review, testing, and release process across teams.

Software Development Team Lead, Codescalers, Jan 2020 - Jan 2023

- Led development of the Jumpscale cloud automation framework, the 3Bot virtual system administrator, and the Virtual Data Center (VDC) platform.
- Drove system architecture and hands-on development while owning code quality and on-time delivery for the team.

Software Engineer, Codescalers, Jan 2016 - Jan 2020

- Designed and built cloud tooling for VM lifecycle management, container orchestration, and infrastructure automation.

Earlier (freelance)

- *Otlob.com* — **Data Analyst**: built an audio-fingerprinting pipeline over large-scale video datasets to measure marketing campaign reach.
- *Orascom Telecom* — **Web Developer**: built an internal PHP/CodeIgniter application for server-room environmental monitoring, replacing manual Excel workflows.

Selected Projects

Operating Systems & Core Infrastructure

- [Zero-OS](#) — lead development of the custom Linux-based OS powering the entire ThreeFold Grid (Go).
- [ContentHive](#) — content-addressed storage system inspired by IPFS (Go).
- [0-Disk](#) — Network Block Device server with pluggable storage backends (Tarantool, Lua).
- [OpenVCloud](#) — monitoring, alerting, and hardware health-checks for a libvirt-based cloud platform (Python).

Platforms & Developer Tools

- [KubeCloud](#) — highly available Kubernetes platform on the ThreeFold Grid (Go, Vue.js, PostgreSQL).
- [VDC](#) — Virtual Data Center: managed Kubernetes clusters, S3 storage, Prometheus/Grafana monitoring (Python, Stellar, Ansible), plus a one-click application marketplace on top of it.
- [AIBroker](#) — AI gateway aggregating multiple LLM providers with MCP server integration (Rust, Go).
- [Terraform Provider](#) — Infrastructure-as-Code plugin for provisioning on the ThreeFold Grid (Go).
- [TFRobot](#) — mass-deployment tool for large VM fleets; [Gridify](#) — Heroku-like deploy tool; [Farmerbot](#) — power-management and operations automation for grid farmers (Go).

SDKs & Frameworks

- [Go SDK](#) — primary SDK enabling delivery of multiple grid-based projects (Go).
- [TypeScript SDK](#) — web developer SDK for ThreeFold Grid integration (TypeScript).
- [JS-NG / JS-SDK](#) — cloud automation framework and Threebot application server (Python, Gevent).
- [EWF](#) — embeddable workflow framework for resilient, stateful multi-step processes (Go).
- [CRM](#) — CRM system for the ThreeFold Token Foundation (Python, Flask, GraphQL, PostgreSQL).

- [publishingtools](#) — web publishing framework for wikis, blogs, and websites (Crystal, Kemal, Svelte).
- [AYS](#) — agentless Infrastructure-as-Code orchestration: event system, background tasks, and service templates.
- [Go farmer](#) — cross-platform GUI for farm management (Go, Fyne). Also: OAuth provider for Gogs, DMIDecode parser, GitHub-Gogs sync, and documentation generators for Jumpscale.

Open Source

- **Networking & infrastructure** — [tcprouter](#): TCP router with SNI-based routing (Go) · [tfgateway](#): ThreeFold gateway built on tcprouter and the CoreDNS Redis plugin · [coredns-redis](#): Redis plugin for CoreDNS · [raqueeb](#): PID 1 replacement and advanced alternative to Foreman.
- **Python libraries** — [pygundb](#): GUN.js database server and client · [pyparsec](#): parser combinator library · [python-sonic-client](#): client for the Sonic search backend · [python-pcre](#): PCRE bindings via ctypes · [flask_itsyouonline](#): OAuth2 middleware for Flask.
- **Nim ecosystem** — [nim-servy](#): web framework · [nim-redisclient](#): Redis client and parser · [nim-assets](#): bundle assets into a single Nim binary.
- **Tools & applications** — [vscode-capnp](#): Cap'n Proto extension for VS Code · [helloED](#): pluggable, fast PyGTK text editor · [stellargui](#): lightweight Stellar blockchain wallet · [raml2spore](#): generate SPORE clients from RAML specifications · [Pida IDE](#): contributed the RegexToolkit and Pastebin plugins.

Publications & Translations

- [Nim Days](#) — book on practical applications and libraries in Nim.
- [Python: Extensive Guide](#) — Arabic programming guide.
- [Introduction to Ruby](#) — Arabic programming guide.
- [Translations](#) — PyQt4, PyGTK, Mono IronPython, and Cairo tutorials translated into Arabic.

Skills

- **Leadership** — team management, hiring, mentoring, technical strategy, roadmap and delivery ownership, architecture review.
- **Languages** — Go, Python, Rust, TypeScript, C, Nim, Crystal, Bash.
- **Infrastructure** — Linux, Kubernetes, K3s, Helm, Terraform/OpenTofu, Ansible, Docker, Prometheus, Grafana, CoreDNS, Nginx, Traefik.
- **Data** — PostgreSQL, Redis, MySQL, MongoDB, InfluxDB.
- **APIs & protocols** — REST, GraphQL, gRPC, Cap'n Proto.
- **Frontend** — Vue, Svelte, React.

Education & Languages

- **BSc in Electrical Engineering** — Helwan University, Cairo, Egypt.
- **Languages** — Arabic (native), English (business proficiency).